List with Header Node

List with Header Node

N S Kumar
Professor
Department of Computer Science &
Engineering (AI & ML)

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 1: List

The list_t has a field called count_ which tracks the number of components in the list.

This implies:

a) This field should be updated on insertion into and deletion of a component from the list

b) The # of operations to find the number of components will still require traversal of the whole list

c) This change is not allowed as it affect the client.

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 1: List

The list_t has a field called count_ which tracks the number of components in the list.

This implies:

a) This field should be updated on insertion into and deletion of a component from the list

b) The # of operations to find the number of components will still require traversal of the whole list

c) This change is not allowed as it affects the client.

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 2: List

Adding at the end of the list

a) May require a few operations not proportional to the length of the list

b) Always requires a number of operations proportional to the length of the list

c) Can never support such an operation

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 2: List

Adding at the end of the list

a) May require a few operations not proportional to the length of the list

b) Always requires a number of operations proportional to the length of the list

c) Can never support such an operation

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 3: List

void what(list_t *ptr_list) { free(ptr_list->head_); }

a) This will free all the nodes in the list

b) This will free all but the first node in the list

c) This will free the first node; rest of the list is accessible

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 3: List

void what(list_t *ptr_list) { free(ptr_list->head_); }

a) This will free all the nodes in the list

b) This will free all but the first node in the list

c) This will free the first node; rest of the list is accessible

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 4: List

The header node is a dummy node added in the beginning of the list.

a) This makes insertion simpler

b) This makes deletion simpler

c) This makes searching simpler

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 4: List

The header node is a dummy node added in the beginning of the list.

a) This makes insertion simpler

b) This makes deletion simpler

c) This makes searching simpler

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 5: List

The tailer node is a dummy node added at the end of the list.

a) This makes insertion simpler

b) This makes deletion simpler

c) This makes searching simpler

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

Quiz 5: List

The tailer node is a dummy node added at the end of the list.

a) This makes insertion simpler

b) This makes deletion simpler

c) This makes searching simpler

d) None of these

© Confidential Information of PES University. All Rights Reserved - 2026

List with Header Node

List

- Is a data structure
- Is an abstract data type
- Has components at an ordinal position
- Two points to note
  - Sequential access
  - Insertion / deletion requires a few operations
- Can implement a list
  - Using linked list
  - Using an array – not discussed
- List can be ordered based on some key

© Confidential Information of PES University. All Rights Reserved - 2026

List with Header Node

List

- Can we change the list implementation without changing the list interface?
- Can we modify the list without affecting the client?
- Add a dummy node in the beginning
- Called the header node
- Will make the insertion simple

© Confidential Information of PES University. All Rights Reserved - 2026

List with Header Node

List

- Initially the head of the list will point to a dummy node
- What should be the value of the key_?

© Confidential Information of PES University. All Rights Reserved - 2026

List with Header Node

List

- Display function will skip over the first and then display the remaining nodes

© Confidential Information of PES University. All Rights Reserved - 2026

List with Header Node

List

- Freeing the list will remove all but the first node

© Confidential Information of PES University. All Rights Reserved - 2026

List with Header Node

List

- Insertion based on the key
  - No empty list ; head_ does not have to change

© Confidential Information of PES University. All Rights Reserved - 2026

List with Header Node

List

- Insertion based on the key
  - No insertion in the beginning; head_ does not have to change

© Confidential Information of PES University. All Rights Reserved - 2026

List with Header Node

List

- Insertion based on the key
  - Insertion in the middle or the end have exactly the same code
  - No selection required once position found

© Confidential Information of PES University. All Rights Reserved - 2026

List with Header Node

List

- With the header node, insertion code has become simpler
- Client is not affected
- Implementation changed independent of the client

© Confidential Information of PES University. All Rights Reserved - 2026

THANK YOU

Professor

Department of Computer Science & Engineering (AI & ML)